

① What is C language?

C language is a programming language that helps you write instructions for the computer to perform tasks. It is fast, powerful, and commonly used to build operating systems, compilers, and many applications.

Eg: • Game Development.

- Compilers and interpreters
- Embedded systems

② Applications of C programming?

C programming is used in operating systems, embedded systems, system software, device drivers, databases, games, and network applications.

- operating systems development (Unix, Linux ports are written in C).
- database systems (MySQL is written in C).
- system software (compilers, assemblers, interpreters).

③ What is variable

A variable is a container used to store data in a program.

Eg: C

```
int age = 20;
```

! (age "b.c" = age "17" h419

Q1) What are different data types in C programming?

The data types in C are int, float, double, char, void, enum, and derived types like arrays, pointers, structures, unions, and functions.

Basic (Primary) data types:

1. int - Integer values.
2. float - decimal values.
3. double - large decimal values.
4. char - single character.

Derived data types:

1. Array
2. Pointer
3. Structure
4. Union
5. Function

Enumeration data type.

enum - used to assign names to integer constants.

void data type.

void - represents "no value" or "empty".

Q2) What is format specifier?

Format specifiers are codes used to represent the type of data in input and output functions.

Eg: C

```
printf("Age = %d", age);
```